SPAST-SENSE: A Human-Centered Wearable Robotic Exoskeleton for Objective Assessment of Post-Stroke Spasticity

Massimo Cenciarini[1], Laia Yébenes-Serra[1], Josep M. Font-Llagunes[1]

[1] Biomechanical Engineering Lab, Department of Mechanical Engineering and Institute for Research and Innovation in Health, Universitat Politècnica de Catalunya – BarcelonaTech (UPC), Spain,
massimo.cenciarini@upc.edu

Introduction

Spasticity is a frequent consequence of stroke that can restrict mobility, interfere with daily activities, and complicate rehabilitation. Its clinical assessment is commonly based on ordinal scales that are practical, but examiner-dependent and unable to provide detailed information on the biomechanical and neural mechanisms underlying resistance to passive movement [1,2]. Robotic-based technologies can be used to apply repeatable movements to the body (passive perturbations) and synchronously measure the resulting response; however, heterogeneity in protocols, devices, and outcome measures continues to limit standardization and eventually clinical translation [3]. The project SPAST-SENSE aims to close this gap by testing the viability of assessing post-stroke spasticity with a wearable and clinically usable knee exoskeleton.

Methods

Building on an existing laboratory prototype of the exoskeleton, the project combines human-centered mechanical redesign, sensor integration, safe control, biomechanical modeling, and clinical validation for developing the proposed SPAST-SENSE platform. The exoskeleton will impose controlled passive knee flexion-extension movements following predefined profiles while recording joint angle, angular velocity, and interaction torque, together with complementary physiological signals where appropriate. Feature extraction, system identification, and data-driven estimation methods will be investigated to derive clinically interpretable spasticity-related indices from the measured torque-angle response. Clinicians and end-users will participate in the design process to guide improvements in ergonomics, comfort, adjustability, portability, and workflow compatibility. Validation will progress from benchtop characterization and laboratory testing to evaluation with post-stroke survivors in collaboration with the Institut Guttmann. Robotic outcomes will be compared with established clinical assessments, while usability and acceptability will be evaluated among clinicians and participants.

Results

The project framework links three components: controlled mechanical perturbations, synchronized sensor measurements, and estimation of a quantitative spasticity index (Figure 1). Expected outputs include the wearable prototype, standardized assessment procedures, a multimodal dataset, and transparent algorithms for extracting biomechanical indicators of hyper-resistance [2]. Together, these outputs will establish the technical and methodological basis for subsequent clinical evaluation and refinement.

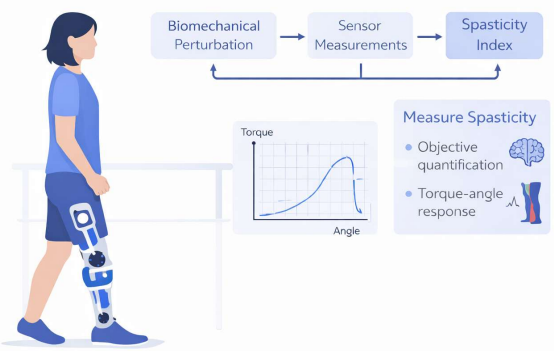


Figure 1: Conceptual framework of SPAST-SENSE. Controlled biomechanical perturbations and synchronized sensor measurements are processed to obtain objective spasticity-related indices.

Conclusions

SPAST-SENSE proposes a translational pathway from laboratory rehabilitation wearable robotics to clinically meaningful biomechanical assessment. The system is intended to complement, rather than replace, clinical judgement by providing repeatable quantitative information for longitudinal monitoring, comparison across evaluators and centers, and more individualized rehabilitation decisions. The resulting platform may also support future adaptations for other neurological conditions associated with spasticity.

Acknowledgements

This work is supported by the Nexus Social 2026 programme of the Universitat Politècnica de Catalunya – BarcelonaTech, the Fundació Caixa Enginyers, and the Impact Hub Barcelona and additionally by the FI STEP predoctoral grant programme (2025 STEP 00551) of the Department of Research and Universities of the Generalitat de Catalunya, co-funded by the European Social Fund Plus.

References

1. Lance JW, Spasticity: Disordered Motor Control, 185-204, 1980.
2. van den Noort et al., Eur J Neurol, 24:981-e38, 2017.
3. Guo et al., J Neuroeng Rehabil, 19:138, 2022.